ILLNESS TREND FORECASTING AND OUTBREAK DETECTION FOR HOSPITALS

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Dissertation submitted in partial fulfillment of the requirements for the Bachelor of Science (Hons) in Information Technology

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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(Mr. Kaniska Yapa)

ABSTRACT

Efficient utilization of hospital resources remains a major challenge due to increasing patient demand, limited infrastructure, and unpredictable admission patterns, often leading to overcrowding and inefficiencies. This study proposes a machine learning–based hospital resource optimization and demand forecasting system to enable proactive and intelligent healthcare management. The framework integrates four components: patient demand forecasting with bed optimization, staff attendance and scheduling analysis, medicine demand forecasting with accident surge detection, and illness trend forecasting using environmental data. The system applies models such as Temporal Fusion Transformers, Random Forest, and XGBoost on historical hospital and environmental datasets to capture complex patterns and support decision-making. Experimental results demonstrate improved forecasting accuracy and resource utilization, reducing waiting times and enhancing operational efficiency. The proposed approach provides a scalable, data-driven solution for effective hospital resource management

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My special appreciation is extended to Yakgahapitiya Hospital, Kandy, for providing the dataset required for this research. The availability of real-world data significantly enhanced the practical value and reliability of this study, allowing the system to be developed and evaluated in a realistic healthcare context.

I would also like to express my gratitude to my group members for their cooperation, teamwork, and mutual support throughout the project. Their collaboration and shared efforts contributed to the successful integration of all system components.

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Table of Contents

DECLARATION iii
ABSTRACT iv
ACKNOWLEDGMENT v
LIST OF ABBREVIATIONS ix
1 INTRODUCTION 11
1.1 General Introduction 11
1.2 Background literature 12
1.3 Research gap 14
1.4 Research Problem 16
1.5 Research Objectives 18
1.5.1 Main Objectives 19
1.5.2 Specific Objective 19
1. Data Collection and Preparation 19
2. Analysis of Disease Trends 20
3. Development of Predictive Model 20
4. Disease Trend Forecasting 20
5. Outbreak Detection Mechanism 20
6. System Development and Visualization 21
7. Model Evaluation and Performance Analysis 21
8. Contribution to Hospital Resource Optimization 21
2 METHODOLOGY 22
2.1 Introduction 22
2.2 System Overview 24
2.3 Data Collection 26
2.5 Feature Engineering 31
Month Feature 31
Week Feature 32
Environmental Features (Temperature and Humidity) 32
Importance of Feature Engineering 32

Final Feature Set 33
2.6 Model Development 33
Random Forest Model 33
Long Short-Term Memory (LSTM) 34
SARIMA Model 34
Prophet Model 35
Model Comparison and Selection 35
Final Model Integration 36
2.7 Model Training and Testing 36
Train-Test Split 36
Training Process 37
Testing Process 37
Evaluation Metrics 37
Mean Absolute Error (MAE) 37
Root Mean Square Error (RMSE) 38
Accuracy 38
Model Comparison and Selection 38
2.8 System Implementation 39
Backend Implementation (Node.js) 39
Frontend Implementation (React) 40
Database Implementation (MongoDB) 40
System Integration 41
Technology Stack Summary 41
3. Commercialization Aspects 42
3.2.1 Introduction 42
3.2.2 Target Users 43
4: RESULTS AND DISCUSSION 51
3 CONCLUSION 62
REFERENCES 63
APPENDICES 65

LIST OF FIGURE

Figure 1 : Overview of Hospital Resource Optimization and Demand Forecasting System 14
Figure 2 Research Objectives Framework 22
Figure 3 Overall System Architecture 26
Figure 4 yakgahapitiya hospital 28
Figure 5 Data Preprocessing Pipeline 30
Figure 6 Prediction Results Actual vs Prediction 51

LIST OF TABLES

Table 1 Identified Research Gaps in Existing Healthcare Forecasting Systems 15
Table 2 Identified Research Gaps in Existing Healthcare Forecasting Systems 17
Table 3 Performance Comparison for Illness Trend Forecasting 39
Table 4 Technology Stack Summary 42

LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
RF	Random Forest
LSTM	Long Short-Term Memory
SARIMA	Seasonal AutoRegressive Integrated Moving Average
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
ML	Machine Learning
AI	Artificial Intelligence
API	Application Programming Interface
UI	User Interface

LIST OF APPENDICES

Appendix	Description	Page
Appendix A	Frontend System Interface	65
Appendix B	System Frontend Codes	67
Appendix C	Backend Model Training (Random Forest)	68

1 INTRODUCTION

1.1 General Introduction

The healthcare sector plays a vital role in ensuring the well-being of society, yet it faces significant challenges in managing resources effectively due to the increasing demand for medical services. Hospitals often experience fluctuations in patient admissions caused by seasonal diseases, epidemics, and other health-related factors. These unpredictable variations make it difficult for hospital administrators to allocate resources such as beds, medical staff, and medicine stock efficiently.

In recent years, the rapid growth of data and advancements in technology have created opportunities to improve healthcare systems through data-driven approaches. Machine learning and data analytics have emerged as powerful tools for analyzing large volumes of healthcare data and identifying patterns that can be used for prediction and decision-making. By leveraging these technologies, it is possible to develop intelligent systems that can forecast future healthcare demands and support proactive resource management.

This research is conducted as part of a broader system titled Hospital Resource Optimization and Demand Forecasting System, which aims to improve the efficiency of hospital operations through predictive analytics. The overall system consists of multiple components designed to address different aspects of hospital management, including resource allocation, demand forecasting, and decision support.

The focus of this individual component is Illness Trend Forecasting and Outbreak Detection, which plays a critical role in predicting disease patterns and identifying potential outbreaks. By analyzing historical disease data, the system forecasts the number of patients likely to be affected by various illnesses in future time periods, such as weekly and monthly intervals. The diseases considered in this study include

Chickenpox, Common Cold, Dengue, Diabetes, Dysentery, Gastritis, Leptospirosis (Rat Fever), Migraine, Respiratory Infections, Typhoid, and Viral Fever.

In addition to forecasting disease trends, this component also incorporates an outbreak detection mechanism. This feature identifies abnormal increases in disease cases and generates early warnings, enabling hospitals to take preventive measures and prepare for potential surges in patient demand. Early detection of outbreaks is essential for reducing the impact on healthcare systems and ensuring timely medical intervention.

To achieve these objectives, a machine learning approach is adopted, utilizing the Random Forest algorithm for prediction. This model is selected due to its high accuracy, robustness, and ability to handle complex datasets. The system is implemented as a web-based application that provides visual representations of predictions, making it easier for hospital staff and decision-makers to interpret results and take appropriate actions.

Overall, this component contributes to the larger goal of optimizing hospital resources by providing accurate and timely predictions of disease trends. By enabling proactive planning and improving decision-making processes, the system enhances the efficiency and effectiveness of healthcare services.

1.2 Background literature

Healthcare systems around the world are continuously challenged by the increasing demand for medical services and limited hospital resources. One of the most critical issues faced by hospitals is the inability to accurately predict patient inflow, especially during periods of disease outbreaks. Seasonal diseases such as dengue, influenza, and viral infections often lead to sudden spikes in hospital admissions, causing overcrowding, resource shortages, and reduced quality of patient care.

Traditionally, hospitals rely on historical trends and manual estimations to manage resources such as hospital beds, medical staff, and medicine stock. However, these conventional methods are often reactive rather than proactive and fail to provide

accurate predictions. As a result, hospitals experience inefficiencies such as underutilized resources during low-demand periods and resource shortages during high-demand periods.

With the advancement of data analytics and machine learning, there has been a growing interest in developing predictive models for healthcare applications. Machine learning algorithms such as Random Forest, Decision Trees, and Neural Networks have been widely used for disease prediction and healthcare forecasting. These models can analyze large volumes of historical data and identify patterns that are not easily visible through traditional methods.

In particular, Random Forest has gained popularity due to its high accuracy, robustness, and ability to handle complex and non-linear datasets. It is an ensemble learning method that constructs multiple decision trees and combines their outputs to improve prediction accuracy. In healthcare applications, Random Forest has been successfully used for disease classification, patient risk prediction, and demand forecasting.

Another important aspect of healthcare forecasting is outbreak detection. Early detection of disease outbreaks allows hospitals and health authorities to take preventive measures, allocate resources efficiently, and reduce the impact on public health. By analyzing trends in disease occurrence, it is possible to identify abnormal increases in cases and generate early warnings.

In this research, a machine learning-based approach is used to forecast illness trends and detect potential outbreaks. The system predicts the number of patients affected by diseases such as Chickenpox, Common Cold, Dengue, Diabetes, Dysentery, Gastritis, Leptospirosis, Migraine, Respiratory Infections, Typhoid, and Viral Fever. These predictions are used to support hospital resource optimization and improve decision-making processes.

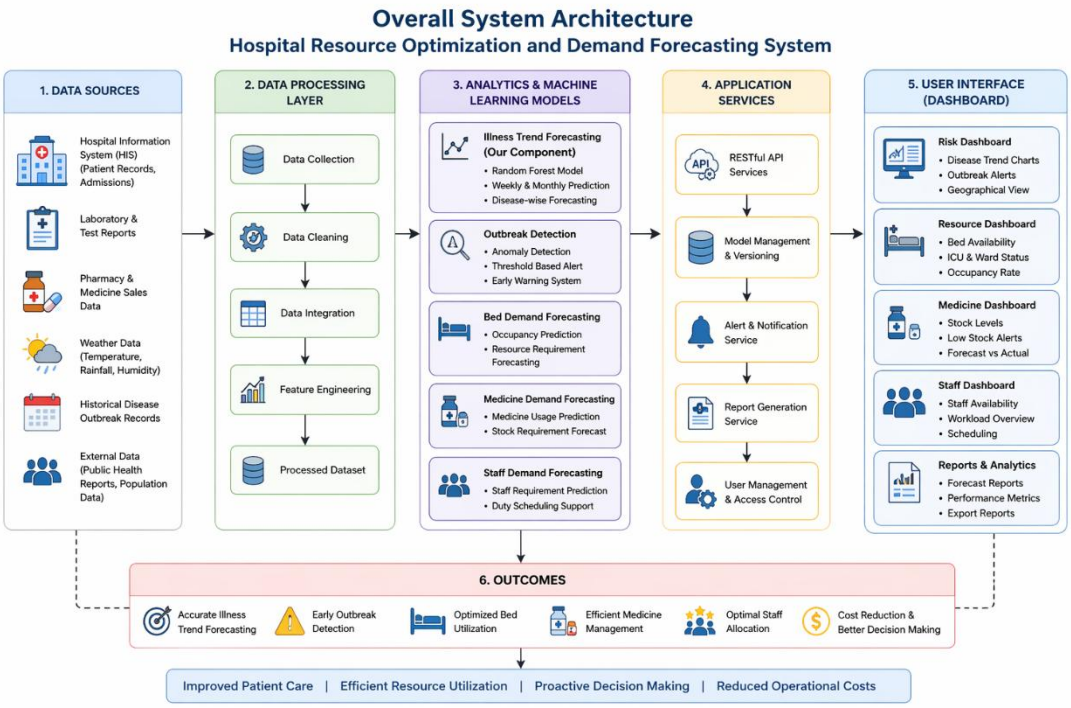


Figure 1 : Overview of Hospital Resource Optimization and Demand Forecasting System

1.3 Research gap

Although many studies have focused on healthcare prediction, several limitations exist in current approaches.

Firstly, most existing systems focus on predicting a single disease rather than multiple diseases. In real-world hospital environments, multiple diseases occur simultaneously, and predicting them individually is not sufficient for effective planning.

Secondly, many models are developed as standalone research prototypes and are not integrated into real-world systems. As a result, their practical usefulness is limited.

Thirdly, outbreak detection is often treated as a separate problem and is not integrated with forecasting models. This lack of integration reduces the effectiveness of prediction systems.

Additionally, there is limited research focused on healthcare systems in developing countries such as Sri Lanka. Most hospitals still rely on manual processes, which are inefficient and prone to errors.

Therefore, there is a clear need for a system that:

- Predicts multiple diseases
- Detects outbreaks
- Supports decision-making
- Integrate into a real system

Table 1 Identified Research Gaps in Existing Healthcare Forecasting Systems

Aspect	Existing Systems	Limitations	Identified Research Gap
Disease Prediction	Focus on single disease	Cannot handle multiple diseases simultaneously	Need multi-disease forecasting system
Forecasting Capability	Limited or no prediction features	Reactive approach based on past data	Need predictive and proactive system
Outbreak Detection	Not integrated with forecasting	Late identification of disease outbreaks	Need early outbreak detection mechanism
System Integration	Standalone models or research prototypes	Not implemented in real-world systems	Need integrated web-based system
Decision Support	Minimal or no visualization tools	Difficult for administrators to interpret data	Need visualization and dashboard support

Data Utilization	Uses limited datasets	Does not consider multiple data sources	Need use of comprehensive healthcare data
Regional Focus	Mostly developed for other countries	Not suitable for local healthcare systems	Need system tailored for Sri Lanka context

1.4 Research Problem

Hospitals face significant challenges in managing resources efficiently due to the unpredictable nature of disease occurrence and patient demand. The number of patients visiting hospitals varies over time as a result of seasonal diseases, environmental conditions, and unexpected outbreaks. This uncertainty makes it difficult for hospital administrators to plan and allocate critical resources such as hospital beds, medical staff, and medicine supplies.

Most existing hospital management systems primarily focus on data storage and administrative functions, lacking the capability to predict future disease trends. As a result, hospitals often rely on manual estimations and historical averages, which are not sufficient to capture complex patterns in healthcare data. This leads to issues such as overcrowding during peak periods, underutilization of resources during low-demand periods, and delays in responding to disease outbreaks.

In addition, early detection of disease outbreaks remains a major challenge. Without an effective outbreak detection mechanism, hospitals are unable to identify abnormal increases in disease cases in a timely manner, resulting in delayed responses and increased risk to public health.

Therefore, the key problem addressed in this research is:

How to develop an accurate and reliable system for forecasting illness trends and detecting potential disease outbreaks in order to support efficient hospital resource optimization and proactive decision-making?

This problem can be further divided into the following sub-problems:

- How to accurately predict the number of patients affected by multiple diseases for future time periods (weekly and monthly)?
- How to detect abnormal increases in disease cases that indicate potential outbreaks?
- How to integrate prediction results into a system that supports hospital planning and decision-making?

One of the key contributors to this issue is the excessive screen time that children are exposed to from a young age. Many preschoolers spend more than the recommended one hour per day on screens, which not only reduces their physical activity but also displaces time that could be spent on play and physical skill. While technology has become an integral part of modern life, including early childhood education, its misuse or overuse can have detrimental effects.

The following conclusions are drawn from an analysis of data from 19,015 preschool children [23].

Table 2 Identified Research Gaps in Existing Healthcare Forecasting Systems

Issue	Description	Impact
Unpredictable Diseases	Sudden increase in cases	Overcrowding
No Forecasting	No future prediction	Poor planning
Manual Methods	Human estimation	Inaccuracy

No Outbreak Detection	Late response	High risk
Resource Mismanagement	Beds/staff shortage	Reduced care quality

1.5 Research Objectives

In modern healthcare environments, hospitals are required to manage a wide range of resources efficiently while responding to continuously changing patient demands. One of the major challenges faced by healthcare institutions is the inability to accurately anticipate future disease trends and patient inflow. This uncertainty often leads to inefficient resource allocation, overcrowded facilities during peak periods, and underutilization during low-demand periods.

The rapid advancement of data analytics and machine learning technologies has created new opportunities to address these challenges through predictive modeling. By analyzing historical healthcare data, it is possible to identify patterns, trends, and correlations that can support accurate forecasting. In particular, the ability to predict disease occurrence and detect potential outbreaks plays a critical role in improving hospital preparedness and decision-making.

This research is conducted as part of the Hospital Resource Optimization and Demand Forecasting System, which aims to enhance the efficiency of healthcare operations through intelligent data-driven solutions. Within this broader system, the Illness Trend Forecasting and Outbreak Detection component focuses on predicting disease patterns and identifying abnormal increases in disease cases.

The research objectives define the direction and scope of this study, guiding the design, implementation, and evaluation of the proposed system. These objectives are

structured into a main objective and a set of specific objectives, each addressing a key aspect of the problem. Together, they ensure that the research systematically addresses the challenges of disease forecasting and outbreak detection while contributing to improved hospital resource management.

1.5.1 Main Objectives

The main objective of this research is To design, develop, and evaluate a machine learning-based system for illness trend forecasting and outbreak detection in order to support efficient hospital resource optimization and enhance data-driven healthcare decision-making.

This objective focuses on creating a comprehensive solution that not only predicts disease trends but also provides actionable insights for hospital administrators. By integrating forecasting and outbreak detection into a single system, the research aims to improve the overall responsiveness and efficiency of healthcare services.

1.5.2 Specific Objective

1. Data Collection and Preparation

To collect relevant historical disease data from available healthcare sources and prepare it for analysis.

This includes gathering data related to different diseases, time periods, and patient counts. The collected data is then cleaned and preprocessed to handle missing values, remove inconsistencies, and ensure data quality. Proper data preparation is essential for building accurate and reliable machine learning models.

2. Analysis of Disease Trends

To analyze historical disease data in order to identify patterns, trends, and seasonal variations.

This step involves exploring how diseases behave over time, including identifying peak periods, recurring patterns, and relationships between different diseases. Understanding these trends provides valuable insights that support the development of effective forecasting models.

3. Development of Predictive Model

To design and implement a machine learning model using the Random Forest algorithm for disease prediction.

Random Forest is selected due to its ability to handle complex datasets, reduce overfitting, and provide high prediction accuracy. The model is trained using historical data and optimized to improve its performance in predicting future disease trends.

4. Disease Trend Forecasting

To predict the number of patients affected by multiple diseases for future time periods, including weekly and monthly forecasts.

This objective focuses on generating accurate predictions that help hospitals anticipate patient demand. By forecasting multiple diseases simultaneously, the system provides a comprehensive view of expected healthcare needs.

5. Outbreak Detection Mechanism

To design and implement an outbreak detection mechanism that identifies abnormal increases in disease cases.

This involves defining threshold values or using anomaly detection techniques to detect sudden spikes in disease occurrence. Early detection of outbreaks allows hospitals to take preventive measures and respond more effectively to emerging health threats.

6. System Development and Visualization

To develop a web-based system that integrates the predictive model and provides visualization of results.

The system is designed to present predictions in a user-friendly format, including graphs, charts, and alerts. This enables hospital administrators and decision-makers to easily interpret the results and make informed decisions.

7. Model Evaluation and Performance Analysis

To evaluate the performance of the developed model using appropriate evaluation metrics such as accuracy, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

This step ensures that the model provides reliable predictions and meets the required performance standards. The evaluation results also help identify areas for improvement and optimization.

8. Contribution to Hospital Resource Optimization

To support hospital resource planning by providing actionable insights based on prediction results.

The system enables hospitals to allocate resources more effectively, reduce operational inefficiencies, and improve patient care. By integrating forecasting with decision-making processes, the research contributes to the overall goal of optimizing healthcare resource utilization.

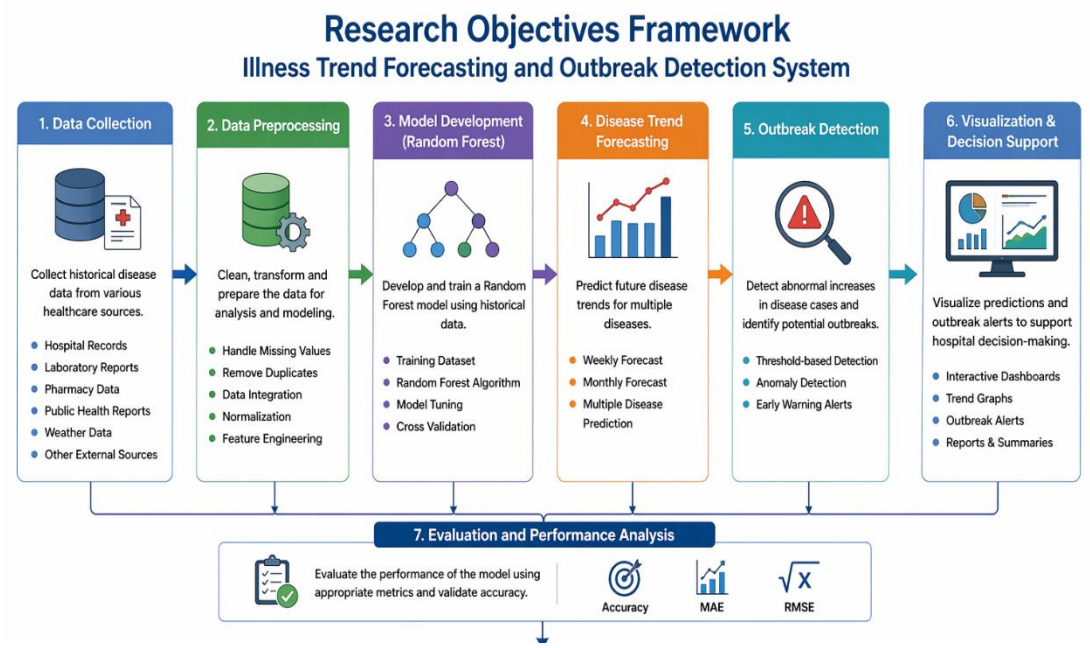


Figure 2 Research Objectives Framework

2 METHODOLOGY

2.1 Introduction

This chapter presents the methodology adopted to design, develop, and evaluate the proposed Illness Trend Forecasting and Outbreak Detection system. The methodology outlines the structured approach used to process healthcare data, apply multiple predictive models, and identify the most effective model for forecasting disease trends.

In modern healthcare systems, accurate prediction of disease patterns is essential for efficient resource management and early outbreak response. However, disease data often exhibits complex temporal patterns, seasonality, and non-linear relationships, making prediction a challenging task. To address these challenges, this research adopts a data-driven approach using both machine learning and time-series forecasting techniques.

The proposed system is developed as part of the broader Hospital Resource Optimization and Demand Forecasting System, where this component focuses specifically on illness trend forecasting and outbreak detection. The methodology is designed to ensure accurate prediction of disease cases and provide reliable insights for hospital decision-making.

The overall methodology consists of several key stages, beginning with data collection and preprocessing. Historical disease data is collected and prepared for analysis by handling missing values, removing inconsistencies, and transforming the data into a suitable format for model training. This step is crucial, as the quality of input data directly affects the performance of predictive models.

Following data preparation, multiple predictive models are implemented to analyze and forecast disease trends. These models include:

- Random Forest (Machine Learning Model)
- Long Short-Term Memory (LSTM) (Deep Learning Model)
- Seasonal AutoRegressive Integrated Moving Average (SARIMA) (Statistical Model)
- Prophet (Time-Series Forecasting Model)

Each model is selected based on its strengths in handling different types of data patterns. LSTM is capable of capturing long-term dependencies in time-series data, SARIMA is effective for seasonal data analysis, and Prophet is designed for handling time-series data with trends and seasonality.

After training the models using historical disease data, their performance is evaluated using appropriate evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Accuracy. This comparative analysis allows the identification of the most suitable model for the given dataset.

Based on the experimental results, the Random Forest model achieved the highest accuracy and best overall performance compared to LSTM, SARIMA, and Prophet.

This indicates that the dataset contains patterns that are effectively captured by ensemble learning methods rather than purely sequential or statistical approaches.

As a result, Random Forest is selected as the final model for deployment in the system. The chosen model is then used to generate predictions for future disease trends, including weekly and monthly forecasts.

In addition to forecasting, an outbreak detection mechanism is incorporated into the system. This mechanism identifies abnormal increases in disease cases by comparing predicted values with expected trends. When a significant deviation is detected, the system generates alerts indicating a potential outbreak, enabling early intervention.

The final stage of the methodology involves integrating the prediction results into a web-based application. The system provides visualization through graphs and dashboards, allowing hospital administrators to interpret results easily and make informed decisions regarding resource allocation.

Overall, this methodology provides a comprehensive framework that combines multiple predictive techniques, comparative evaluation, and practical implementation. By selecting the most accurate model and integrating it into a real-world system, the research contributes to improving hospital resource optimization and enhancing healthcare service delivery.

2.2 System Overview

The proposed system is developed as part of the broader Hospital Resource Optimization and Demand Forecasting System, which aims to enhance hospital efficiency through predictive analytics and intelligent decision-making. The overall system consists of multiple components that work together to analyze healthcare data, forecast future demand, and optimize resource utilization.

Within this system, the Illness Trend Forecasting and Outbreak Detection component plays a critical role by predicting future disease trends and identifying potential outbreaks. This component supports hospital administrators by providing insights into expected patient inflow, enabling proactive planning and efficient allocation of resources such as hospital beds, medical staff, and medicine supplies.

The primary functionality of this component is divided into two main tasks: disease prediction and outbreak detection. The system analyzes historical disease data and uses machine learning techniques to forecast the number of patients affected by various

diseases for future time periods, including weekly and monthly intervals. The diseases considered in this research include Dengue, Chickenpox, Common Cold, Diabetes, Dysentery, Gastritis, Leptospirosis, Migraine, Respiratory Infections, Typhoid, and Viral Fever.

In addition to forecasting disease trends, the system incorporates an outbreak detection mechanism. This mechanism monitors predicted values and identifies abnormal increases in disease cases. When the predicted number of cases exceeds a predefined threshold or deviates significantly from expected patterns, the system generates alerts indicating a potential outbreak. This early warning capability allows healthcare providers to take preventive measures and minimize the impact of disease spread.

The overall system architecture is illustrated in Figure 2.1, which shows the flow of data and the interaction between different components of the system. The architecture consists of several key layers, including data input, data preprocessing, model processing, prediction generation, outbreak detection, and visualization.

At the initial stage, the system collects historical disease data from relevant sources. This data includes attributes such as date, disease type, and patient count. The collected data is then passed through the preprocessing stage, where it is cleaned, transformed, and prepared for model training.

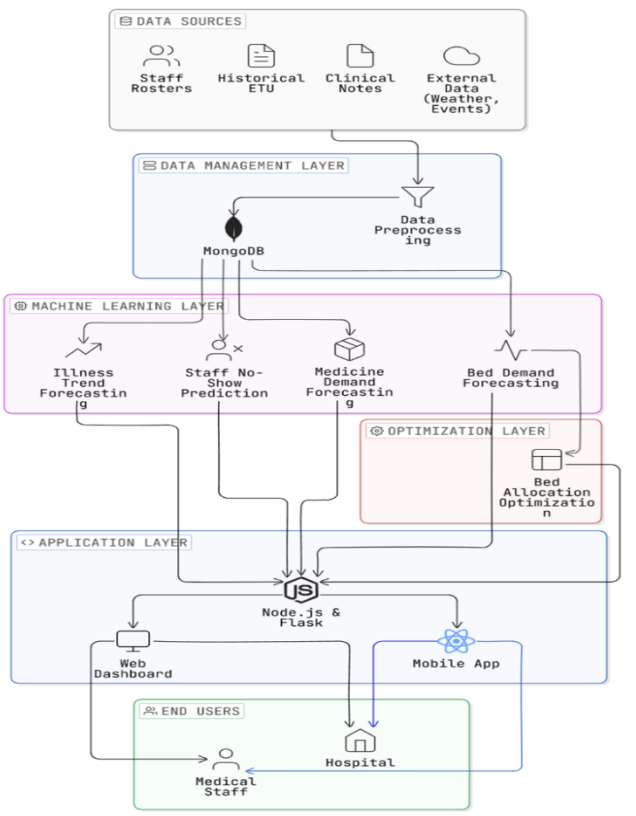
Following preprocessing, the processed data is fed into the model layer, where multiple predictive models are applied. These include Random Forest, LSTM, SARIMA, and Prophet models. Each model analyzes the data and generates predictions for future disease trends. The performance of these models is evaluated, and the best-performing model, which in this case is the Random Forest model, is selected for final prediction.

The prediction results are then passed to the outbreak detection module. This module compares the predicted values with predefined thresholds or historical patterns to identify abnormal increases in disease cases. If an anomaly is detected, the system triggers an alert indicating a potential outbreak.

Finally, the results are displayed through a web-based interface, where predictions and alerts are visualized using graphs and charts. This visualization enables hospital administrators to easily interpret the results and make informed decisions regarding resource allocation and planning.

Overall, the system provides a comprehensive solution for illness trend forecasting and outbreak detection. By integrating data processing, machine learning, and visualization, the system enhances the ability of hospitals to respond effectively to changing healthcare demands.

Figure 3 Overall System Architecture



2.3 Data Collection

Data collection is a fundamental step in the development of the proposed illness trend forecasting and outbreak detection system, as the accuracy and reliability of the predictive models depend heavily on the quality of the dataset. In this research, real-world healthcare data was obtained from Yakgahapitiya Hospital, Kandy, providing a practical and realistic foundation for the study.

The dataset consists of historical disease records collected over a period of ten years, from 2016 to 2025. These records represent monthly observations of patient counts for multiple diseases reported at the hospital. The use of real hospital data ensures that the developed system reflects actual healthcare conditions and improves the applicability of the results in real-world scenarios.

The dataset includes information on 11 different diseases, namely Dengue, Chickenpox, Common Cold, Diabetes, Dysentery, Gastritis, Leptospirosis (Rat

Fever), Migraine, Respiratory Infections, Typhoid, and Viral Fever. Each disease contains approximately 120 records, corresponding to monthly data collected over the 10-year period. In total, the dataset comprises approximately 1,320 observations, which provides a sufficient volume of data for training and evaluating machine learning models.

In addition to disease case counts, the dataset incorporates environmental factors, specifically temperature and humidity, which are known to influence the spread and occurrence of various diseases. These environmental variables enhance the predictive capability of the model by enabling it to capture relationships between climatic conditions and disease trends.

Each record in the dataset includes key attributes such as:

- Date (month and year)
- Disease name
- Patient count
- Temperature
- Humidity

The dataset is organized in a structured tabular format, where each row represents a specific disease observation for a given time period. This structure makes the dataset suitable for both machine learning models, such as Random Forest, and time-series models, including LSTM, SARIMA, and Prophet.

Before using the dataset for model training, ethical considerations were taken into account to ensure that no personally identifiable patient information was included. The dataset only contains aggregated disease counts, ensuring compliance with data privacy and confidentiality requirements.

Overall, the use of real-world data from Yakgahapitiya Hospital significantly enhances the reliability and validity of this research. It enables the development of a practical system that can be applied in real healthcare environments for illness trend forecasting and outbreak detect

Figure 4 yakgahapitiya hospital

ion.

2.4 Data Preprocessing

Data preprocessing is a critical stage in the development of any machine learning-based system, as the quality of input data directly influences the accuracy and reliability of the predictive model. In this research, the collected dataset undergoes several preprocessing steps to ensure consistency, completeness, and suitability for model training.The dataset consists of time-series records related to multiple diseases and environmental factors. Since raw data may contain inconsistencies, missing values, and noise, it is necessary to perform preprocessing to transform the data into a structured and clean format.

The preprocessing stage includes several key steps such as data cleaning, handling missing values, data encoding, and normalization, which are described below.

1. Data Cleaning

   Data cleaning is the initial step in preprocessing, where inconsistencies and irrelevant information are removed from the dataset. In this study, the dataset is examined for duplicate records, incorrect values, and formatting errors.Duplicate entries are identified and removed to avoid bias in model training. In addition, inconsistent data formats, such as incorrect date formats or invalid numerical values, are corrected. Ensuring clean and consistent data is essential for improving the performance of predictive models.

2. Handling Missing Values

   Real-world datasets often contain missing or incomplete data. Missing values can negatively affect model accuracy if not handled properly. Therefore, appropriate techniques are applied to manage missing data.In this research, missing values are handled using methods such as:

   Mean or median imputation for numerical attributes (e.g., temperature, humidity)

   Forward filling or interpolation for time-series data

   Removing records with excessive missing values if necessary

   These methods ensure that the dataset remains complete while preserving the underlying data patterns.

3. Data Encoding

   Machine learning models require numerical input data. Therefore, categorical variables such as disease names must be converted into numerical representations.In this study, label encoding or one-hot encoding techniques are used to transform categorical variables into numerical form. For example, disease names are encoded into numerical values, allowing the model to process them effectively.

   Encoding ensures compatibility between the dataset and the machine learning algorithms used in this research.

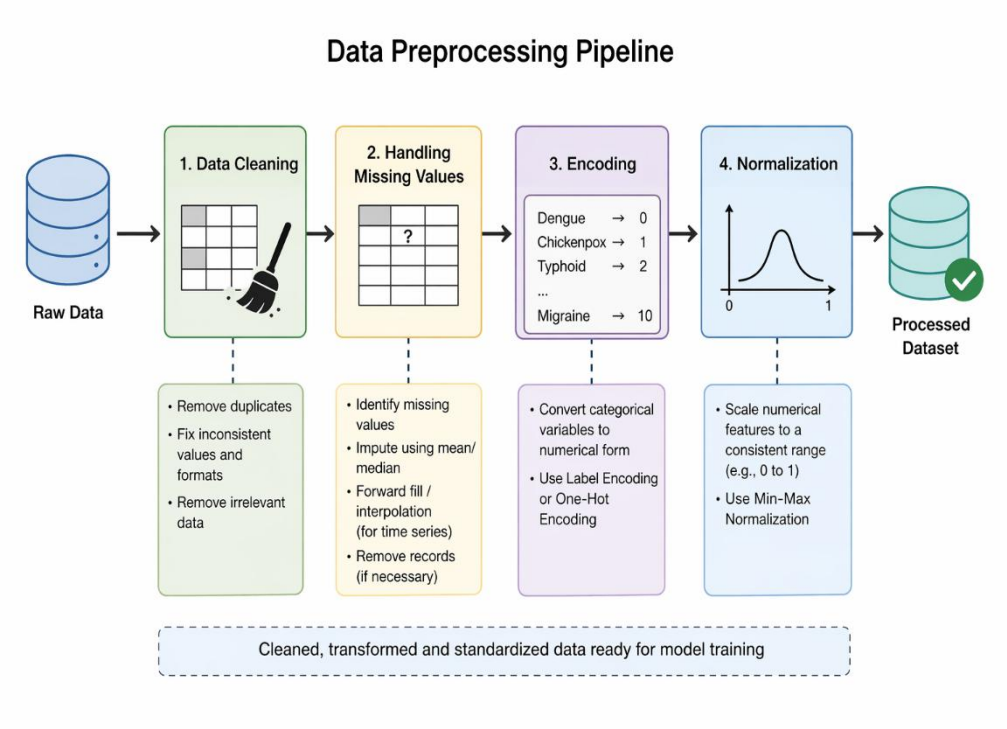
4. Normalization

Normalization is applied to scale numerical features into a consistent range, typically between 0 and 1. This step is important for improving model performance and ensuring that no single feature dominates the prediction process.In this research, normalization is applied to environmental variables such as rainfall, temperature, and humidity. Scaling these values helps the model learn patterns more effectively and improves prediction accuracy.

5. Final Prepared Dataset

After completing all preprocessing steps, the dataset is transformed into a clean, structured, and machine-learning-ready format. The processed dataset contains consistent and normalized values, making it suitable for training predictive models such as Random Forest, LSTM, SARIMA, and Prophet. Overall, the preprocessing stage plays a crucial role in enhancing the quality of the dataset and ensuring the effectiveness of the proposed illness trend forecasting and outbreak detection system.

Figure 5 Data Preprocessing Pipeline



2.5 Feature Engineering

Feature engineering is a crucial step in the machine learning pipeline, as it involves transforming raw data into meaningful features that improve the performance of predictive models. In the context of healthcare data, particularly time-series disease data, feature engineering plays a significant role in capturing temporal patterns and relationships that influence disease occurrence.

The dataset used in this research consists of time-based observations, where disease counts are recorded on a monthly basis along with environmental factors. Since machine learning models such as Random Forest do not inherently understand time dependencies, it is necessary to extract and construct relevant features that represent temporal information effectively.

In this study, several time-based features are engineered, including month, week, and seasonal patterns, to enhance the predictive capability of the models.

Month Feature

The month feature is extracted from the date attribute and represents the specific month (1–12) in which the disease data is recorded. This feature is important because many diseases exhibit monthly variations due to environmental and climatic conditions.

For example, diseases such as dengue and respiratory infections often show increased cases during certain months associated with rainfall or temperature changes. By including the month as a feature, the model can learn these recurring patterns and improve its forecasting accuracy.

Week Feature

In addition to monthly data, a week-based representation can be derived to capture finer temporal patterns. The week feature represents the week number within a year (1–52) and provides a more granular view of disease trends.

Although the primary dataset is monthly, converting time into weekly representation helps improve the model's ability to detect short-term fluctuations and sudden changes in disease patterns. This is particularly useful for outbreak detection, where early identification of abnormal increases is critical.

Environmental Features (Temperature and Humidity)

Environmental factors such as temperature and humidity play a significant role in disease transmission and spread. For example:

- High humidity levels can increase the survival rate of pathogens and vectors such as mosquitoes
- Temperature variations can influence the spread of respiratory infections

These features are directly included in the dataset and used by the model to learn relationships between environmental conditions and disease occurrence. Incorporating these variables enhances the model's ability to capture real-world patterns and improves prediction accuracy.

Importance of Feature Engineering

The inclusion of time-based features such as month, week, and seasonal indicators allows the model to capture both short-term and long-term patterns in disease data. These features enhance the model's ability to learn complex relationships between time, environmental factors, and disease occurrence.

Feature engineering also helps improve the performance of machine learning models by providing relevant and informative inputs. In this research, the engineered features significantly contribute to the effectiveness of the Random Forest model, enabling it to achieve higher accuracy compared to other models.

Final Feature Set

After feature engineering, the dataset includes both original and derived features, such as:

- Date-based features (Month, Week)
- Environmental features (Rainfall, Temperature, Humidity)
- Disease-related features (Patient Count)
- Seasonal indicators

2.6 Model Development

In order to accurately forecast disease trends and detect potential outbreaks, multiple predictive models were developed and evaluated in this research. Since the dataset contains time-based disease records along with environmental factors such as temperature and humidity, it is important to consider both machine learning and time-series forecasting techniques.

The models implemented in this study include Random Forest, Long Short-Term Memory (LSTM), Seasonal AutoRegressive Integrated Moving Average (SARIMA), and Prophet. Each model is selected based on its ability to capture different characteristics of the dataset, such as non-linear relationships, temporal dependencies, and seasonal patterns.

Random Forest Model

Random Forest is an ensemble machine learning algorithm that builds multiple decision trees and combines their outputs to produce a final prediction. Each decision tree is trained using a random subset of the data and features, which improves the robustness and generalization capability of the model.

In this research, Random Forest is used to model the relationship between disease occurrence and influencing factors such as time (month, week) and environmental variables (temperature and humidity). The model splits the dataset into multiple decision paths based on feature values and aggregates the results to generate accurate predictions.

The key advantages of Random Forest include:

- Ability to handle non-linear relationships in the data
- High resistance to overfitting due to ensemble learning
- Capability to work effectively with both numerical and categorical features

Due to these characteristics, Random Forest demonstrated superior performance in capturing complex disease patterns and achieved the highest prediction accuracy among all models used in this study.

Long Short-Term Memory (LSTM)

LSTM is a type of deep learning model based on recurrent neural networks (RNNs), specifically designed to process sequential and time-series data. It is capable of learning long-term dependencies through memory cells and gating mechanisms.

In this research, LSTM is applied to analyze temporal patterns in disease data by considering sequences of past observations to predict future values. This makes it suitable for capturing trends and dependencies over time. However, LSTM requires a large amount of data and computational resources for effective training. In this study, although LSTM was able to capture temporal patterns, its performance was moderate compared to Random Forest due to the relatively limited dataset size and complexity.

SARIMA Model

SARIMA (Seasonal AutoRegressive Integrated Moving Average) is a statistical model used for time-series forecasting, particularly for data exhibiting seasonality and trends. It extends the ARIMA model by incorporating seasonal components.

In this study, SARIMA is used to model the seasonal behavior of disease occurrences. It analyzes relationships between current values and past observations,

taking into account seasonal variations. While SARIMA is effective for linear time-series data, it has limitations in handling complex and non-linear patterns. As a result, its prediction accuracy in this research was lower compared to machine learning models such as Random Forest.

Prophet Model

Prophet is a time-series forecasting model developed by Facebook, designed to handle data with strong trends and seasonal patterns. It is particularly useful for datasets with missing values and irregular intervals.

In this research, Prophet is used to model disease trends by decomposing the time series into trend, seasonality, and residual components. It provides an easy-to-use framework for forecasting and performs well in capturing general patterns.However, Prophet assumes relatively simple relationships within the data and may not perform well when dealing with complex interactions between multiple variables. Therefore, its performance in this study was moderate compared to Random Forest.

Model Comparison and Selection

After implementing all four models, a comparative evaluation was conducted using performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Accuracy.

The results indicate that:

- Random Forest achieved the highest accuracy
- LSTM provided moderate performance
- SARIMA showed lower performance due to linear assumptions
- Prophet performed reasonably but was less accurate for complex patterns

Based on this analysis, Random Forest was selected as the final model for the proposed system. The superior performance of Random Forest can be attributed to its ability to effectively capture non-linear relationships, handle multiple features, and generalize well across different disease patterns.

Final Model Integration

The selected Random Forest model is integrated into the system to generate predictions for future disease trends. The model outputs the expected number of cases for each disease over specified time periods, such as weekly and monthly forecasts. These predictions are then used in the outbreak detection module and visualization system, supporting hospital administrators in making informed decisions regarding resource allocation and planning.

2.7 Model Training and Testing

Model training and testing is a crucial stage in the development of the proposed illness trend forecasting and outbreak detection system. This stage ensures that the predictive models are properly trained using historical data and evaluated for their accuracy and reliability before deployment.

In this research, the dataset obtained from Yakgahapitiya Hospital, Kandy is divided into two subsets: a training dataset and a testing dataset. The purpose of this division is to train the models on one portion of the data and evaluate their performance on unseen data, thereby ensuring that the models generalize well to future predictions.

Train-Test Split

The dataset is split into training and testing sets using a standard ratio, typically 80% for training and 20% for testing. The training dataset is used to train the predictive models, while the testing dataset is used to evaluate model performance.

Since the dataset is time-series based, the split is performed chronologically to preserve the temporal order of data. Earlier records are used for training, and more recent records are used for testing. This approach ensures that the model is evaluated on future data, which closely reflects real-world prediction scenarios.

Training Process

During the training phase, each model (Random Forest, LSTM, SARIMA, and Prophet) is trained using the processed dataset. The models learn patterns and relationships between input features such as month, week, temperature, and humidity, and the target variable, which is the number of disease cases.

- Random Forest learns by constructing multiple decision trees and aggregating their outputs.
- LSTM learns sequential dependencies in the time-series data.
- SARIMA models seasonal patterns using statistical techniques.
- Prophet captures trends and seasonality through decomposition.

The training process involves adjusting model parameters to minimize prediction errors and improve accuracy.

Testing Process

After training, the models are evaluated using the testing dataset. The testing process involves feeding unseen data into the trained models and comparing the predicted values with the actual observed values.

This step is essential to measure how well the models perform on new data and to ensure that they are not overfitting the training dataset.

Evaluation Metrics

To assess the performance of the models, several evaluation metrics are used:

Mean Absolute Error (MAE)

MAE measures the average absolute difference between predicted and actual values. It provides a simple interpretation of prediction error.

Root Mean Square Error (RMSE)

RMSE measures the square root of the average squared differences between predicted and actual values. It gives higher weight to larger errors and is useful for identifying significant prediction deviations.

Accuracy

Accuracy represents the overall correctness of predictions. In this research, it is used as a comparative measure to evaluate the performance of different models.

These metrics provide a comprehensive evaluation of model performance and help identify the most suitable model for disease forecasting.

Model Comparison and Selection

Based on the evaluation results, a comparative analysis is conducted to determine the best-performing model. The results indicate that:

- Random Forest achieved the lowest error values (MAE and RMSE)
- Random Forest demonstrated the highest accuracy
- LSTM provided moderate performance
- SARIMA showed lower performance due to linear assumptions
- Prophet performed reasonably but was less accurate for complex data patterns

Therefore, Random Forest is selected as the final model for the proposed system due to its superior performance and reliability.

Table 3 Performance Comparison for Illness Trend Forecasting

Model	MAE	RMSE	MAPE %
SARIMA	12.45	28.4	18.9
LSTM	8.54	10.20	12.3
Prophet	10.12	12.85	14.7
Random forest (Weekly)	0.82	1.25	6.4
Random forest (Monthly)	3.15	4.10	9.2

2.8 System Implementation

The proposed Illness Trend Forecasting and Outbreak Detection system is implemented as a web-based application using a modern full-stack architecture. The system integrates a React-based frontend, a Node.js backend, and a MongoDB database, enabling efficient data processing, prediction generation, and user interaction.

The implementation is designed to ensure scalability, flexibility, and real-time accessibility, allowing hospital administrators to interact with the system through an intuitive interface and obtain meaningful insights into disease trends and potential outbreaks.

Backend Implementation (Node.js)

The backend of the system is developed using Node.js, which provides a scalable and efficient environment for handling server-side operations. Node.js is chosen due to its non-blocking, event-driven architecture, making it suitable for applications that require real-time data processing.

The backend is responsible for:

- Handling API requests from the frontend
- Processing and managing data

- Integrating machine learning models for prediction
- Performing outbreak detection logic
- Communicating with the database

RESTful APIs are developed to enable communication between the frontend and backend. These APIs allow users to request predictions, retrieve historical data, and view outbreak alerts. The prediction logic, including the trained Random Forest model, is integrated into the backend. When a request is made, the backend processes input data, generates predictions, and returns results to the frontend.

Frontend Implementation (React)

The frontend of the system is developed using React.js, a popular JavaScript library for building user interfaces. React is selected due to its component-based architecture, which allows for efficient development of interactive and dynamic web applications.

The frontend is responsible for:

- Displaying disease prediction results
- Visualizing data using charts and graphs
- Providing user-friendly interfaces for interaction
- Displaying outbreak alerts and notifications

The interface includes dashboards that present:

- Weekly and monthly disease forecasts
- Graphical representation of trends
- Alerts for potential outbreaks

React enables real-time updates and smooth user experience, allowing users to easily interpret prediction results and make informed decisions.

Database Implementation (MongoDB)

The system uses MongoDB as the database for storing and managing data. MongoDB is a NoSQL database that stores data in a flexible, JSON-like format, making it suitable for handling structured and semi-structured data.

The database is responsible for:

- Storing historical disease data
- Managing environmental data (temperature and humidity)
- Saving prediction results and logs
- Supporting data retrieval for analysis

MongoDB is chosen due to its scalability, flexibility, and ability to handle large datasets efficiently. The document-based structure allows easy storage of time-series data and supports quick data access.

System Integration

The integration of the frontend, backend, and database forms a complete system architecture. The workflow of the system is as follows:

- The user interacts with the React frontend
- The frontend sends requests to the Node.js backend via APIs
- The backend processes data and retrieves information from MongoDB
- The prediction model generates disease forecasts
- The backend sends results back to the frontend
- The frontend displays predictions and alerts

This architecture ensures seamless communication between system components and enables efficient processing of healthcare data.

Technology Stack Summary

The combination of React, Node.js, and MongoDB provides several advantages:

- Scalability and flexibility
- Real-time data processing
- Efficient handling of large datasets
- User-friendly interface for visualization

This modern web-based architecture makes the system suitable for deployment in real-world hospital environments.

Table 4 Technology Stack Summary

Component	Technology	Purpose
Frontend	React.js	User interface and visualization
Backend	Node.js	API handling and processing
Database	MongoDB	Data storage and retrieval
Model Integration	Machine Learning Models	Disease prediction and analysis

3. Commercialization Aspects

3.2.1 Introduction

The commercialization aspect of the proposed Illness Trend Forecasting and Outbreak Detection system focuses on how the developed solution can be applied in real-world healthcare environments. While the technical implementation demonstrates the feasibility of predicting disease trends, commercialization ensures

that the system delivers practical value to healthcare institutions, improves operational efficiency, and supports decision-making processes.

The proposed system is designed not only as a research prototype but also as a scalable and deployable solution that can be integrated into hospital management systems. By leveraging predictive analytics, the system provides actionable insights that can significantly enhance hospital resource optimization and healthcare service delivery.

3.2.2 Target Users

Hospitals

Hospitals are the main beneficiaries of the system. The application can be used by hospital administrators, medical officers, and operational managers to forecast disease trends and prepare for patient inflow. By utilizing the system, hospitals can allocate resources such as beds, medical staff, and medicines more efficiently.

Government Health Departments

Government health authorities can use the system to monitor disease trends across regions and identify potential outbreaks at an early stage. This enables better public health planning, policy-making, and resource distribution at a national level.

Public Health Organizations

Organizations involved in disease surveillance and healthcare management can utilize the system for research and monitoring purposes.

3.2.3 Business Value

The proposed system offers significant business and operational value to healthcare institutions:

- **Cost Reduction**

By accurately predicting disease trends, hospitals can avoid unnecessary resource allocation and reduce operational costs. Efficient planning minimizes wastage of medical supplies and optimizes staff utilization.

- **Improved Planning and Decision-Making**
  The system provides predictive insights that enable proactive planning. Hospital administrators can prepare for future demand instead of reacting to sudden increases in patient numbers.
- **Enhanced Patient Care**
  By ensuring the availability of necessary resources, the system improves the quality of patient care. Reduced overcrowding and better preparedness lead to faster treatment and improved patient outcomes.
- **Early Outbreak Response**
  The outbreak detection feature allows hospitals and health authorities to take preventive actions early, reducing the spread and impact of diseases.

3.2.4 Deployment Strategy

The proposed system can be deployed using modern cloud-based technologies to ensure accessibility and scalability.

Cloud-Based Deployment

The system can be hosted on cloud platforms such as AWS, Azure, or Google Cloud. This allows hospitals to access the system remotely without requiring complex infrastructure setup.

Software as a Service (SaaS) Model

The system can be offered as a SaaS solution, where hospitals subscribe to the service and access it through a web interface. This model eliminates the need for local installation and simplifies maintenance and updates.

Integration with Existing Systems

The system can be integrated with existing hospital management systems through APIs, enabling seamless data exchange and improved functionality

3.2.5 Revenue Model

The commercialization of the system can be achieved through various revenue models:

- **Subscription-Based Model**
  Hospitals can pay a monthly or annual subscription fee to access the system. Different pricing tiers can be offered based on features and usage levels.
- **Government Licensing**
  Government health departments can license the system for nationwide deployment. This allows centralized monitoring of disease trends and efficient resource allocation across multiple hospitals.
- **Custom Solutions**
  Customized versions of the system can be developed for specific hospitals or regions, generating additional revenue.

3.2.6 Scalability

The proposed system is designed to be scalable, allowing it to expand and adapt to different requirements:

Expansion to More Diseases

The system can be extended to include additional diseases by incorporating new datasets and retraining the model.

Deployment Across Multiple Hospitals

The system can be implemented in multiple hospitals and healthcare facilities, enabling large-scale data analysis and forecasting.

Integration with Additional Data Sources

Future enhancements can include integration with external data sources such as weather data, population data, and epidemiological reports to improve prediction accuracy.

3.2.7 Challenges and Limitations

Despite its advantages, the commercialization of the system involves several challenges:

- **Data Privacy and Security:** Healthcare data is highly sensitive, and strict measures must be taken to ensure data privacy and compliance with regulations. The system must implement secure data handling and access control mechanisms.

- **Integration with Hospital Systems:** Integrating the system with existing hospital management systems may require technical adjustments and compatibility considerations.

- **Data Availability and Quality:** The effectiveness of the system depends on the availability of high-quality data. Incomplete or inaccurate data may affect prediction accuracy.

- **User Adoption:** Healthcare professionals may require training to effectively use the system. Ensuring user acceptance is important for successful implementation.

3.2.8 Summary

The commercialization aspect of the proposed system demonstrates its potential to be implemented as a practical solution in real-world healthcare environments. By providing predictive insights, improving resource management, and enabling early outbreak detection, the system offers significant value to hospitals and government health authorities.

The use of cloud-based deployment and SaaS models ensures scalability and accessibility, while the revenue models provide opportunities for sustainable

implementation. Although challenges such as data privacy and system integration exist, they can be addressed through proper planning and technological solutions.

Overall, the proposed system has strong potential for commercialization and can contribute to improving healthcare efficiency and patient outcomes.

3.3 Testing and Implementation

3.3.1 System Implementation

The proposed Illness Trend Forecasting and Outbreak Detection system is implemented as a web-based application using a full-stack architecture. The system is developed using React for the frontend, Node.js for the backend, and MongoDB for the database, ensuring scalability, flexibility, and efficient data processing.

The system architecture follows a client-server model, where the frontend interacts with the backend through RESTful APIs. The backend processes requests, interacts with the database, and integrates the machine learning models to generate predictions.

The workflow of the system is as follows:

- Users interact with the React-based frontend interface
- Requests are sent to the Node.js backend
- The backend retrieves data from MongoDB
- The trained model (Random Forest) generates predictions
- The outbreak detection module analyzes prediction results
- The system sends results back to the frontend
- Predictions and alerts are displayed to users

This architecture ensures efficient communication between components and enables real-time prediction and visualization.

3.3.2 Testing Types

To ensure the reliability and effectiveness of the system, multiple types of testing were conducted.

Functional Testing

Functional testing is performed to verify that all system features operate correctly according to the requirements. This includes:

- Verifying data input and retrieval
- Checking prediction generation
- Ensuring correct display of graphs and alerts
- Validating API responses

The results confirm that the system functions as expected and provides accurate outputs for user inputs.

Model Testing

Model testing is conducted to evaluate the accuracy and reliability of the predictive models. Each model (Random Forest, LSTM, SARIMA, and Prophet) is tested using the testing dataset.

The performance is measured using evaluation metrics such as:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- Accuracy

The testing results indicate that the Random Forest model achieved the highest accuracy and lowest error values compared to other models.

Performance Testing

Performance testing is carried out to assess the system's speed and responsiveness. This includes:

- Measuring response time for prediction requests
- Evaluating system performance under multiple requests
- Checking database query efficiency

The results show that the system provides fast response times and can handle user requests efficiently, making it suitable for real-world applications.

3.3.3 Results

The system successfully generates predictions for disease trends based on historical data. The results include:

- Weekly disease forecasts
- Monthly disease predictions
- Graphical visualization of trends
- Outbreak alerts for abnormal increases

The prediction results are displayed using graphs and charts, allowing users to easily interpret the data. These visualizations help identify patterns such as seasonal variations and sudden increases in disease cases.

The system demonstrates the ability to accurately forecast disease trends and detect potential outbreaks, supporting proactive decision-making in healthcare management.

3.3.4 Evaluation

A comparative evaluation of all models is conducted to determine the most effective approach for disease prediction. The results show that:

- Random Forest achieved the best performance
- LSTM provided moderate results

- SARIMA showed lower accuracy due to linear assumptions
- Prophet performed reasonably but was less effective for complex patterns

The superior performance of Random Forest is attributed to its ability to capture non-linear relationships and handle multiple input features effectively.Based on these results, Random Forest is selected as the final model for the system, ensuring high prediction accuracy and reliability.

3.3.5 Limitations

Despite the effectiveness of the proposed system, several limitations are identified:

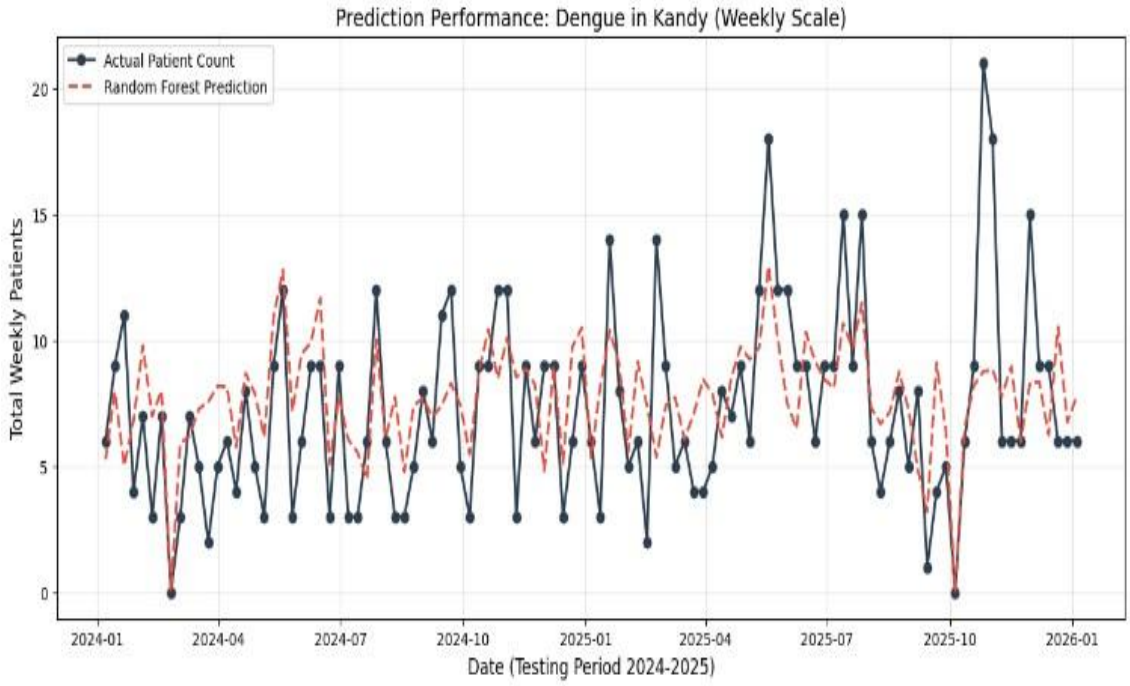
1. Limited Dataset: The dataset used in this research is limited to data collected from a single hospital (Yakgahapitiya Hospital). This may affect the generalizability of the model when applied to other regions.

2. Lack of Real-Time Data: The system relies on historical data and does not currently support real-time data integration. This may limit its ability to respond to sudden changes in disease patterns.

3. Environmental Factors: Only temperature and humidity are considered as environmental factors. Other factors such as rainfall and population density could further improve prediction accuracy.

4. Model Limitations: Although Random Forest performed best, it may not capture long-term temporal dependencies as effectively as deep learning models like LSTM.

Figure 6 Prediction Results Actual vs Prediction



4: RESULTS AND DISCUSSION

4.1 Results

The results of this research demonstrate the successful implementation and effectiveness of the proposed Illness Trend Forecasting and Outbreak Detection system. The system was developed using real-world data obtained from Yakgahapitiya Hospital, Kandy, and tested using multiple predictive models, including Random Forest, LSTM, SARIMA, and Prophet.

The system provides a comprehensive web-based interface that allows users to visualize disease trends, generate forecasts, and identify potential outbreaks. The results are presented through dashboards, graphs, and alert systems, which enhance usability and support decision-making.

to ensure that the pose detection system matched the physical abilities of children and provided helpful feedback. This process can be summarized in the following steps.

4.1.1 Disease Forecasting Results

The system successfully generates monthly and annual forecasts for multiple diseases. The forecast dashboard displays a 12-month prediction trend, allowing users to observe how disease cases are expected to change over time.The results indicate that the system can effectively capture temporal patterns and seasonal variations in disease occurrence. For example, diseases such as dengue show gradual increases during certain months, which may be influenced by environmental conditions such as temperature and humidity.

The forecasting results demonstrate that the model is capable of providing consistent and realistic predictions, which can be used by hospital administrators for planning and resource allocation.

4.1.2 Weekly and Monthly Prediction Outputs

The system provides both short-term and long-term predictions, including:

- Weekly predictions for immediate planning
- Monthly predictions for strategic decision-making

For example, the system predicts:

- Weekly expected admissions (e.g., 8 patients for dengue)
- Monthly expected admissions (e.g., 41 patients for dengue)

These predictions are displayed in a clear and user-friendly format using dashboard cards, enabling users to quickly understand expected patient inflow.

4.1.3 Trend Analysis Results

The system includes a trend analysis module that visualizes historical disease patterns and compares multiple diseases over time.The trend analysis graph shows the behavior of different diseases, such as dengue, viral fever, respiratory infections, and gastritis. The results indicate that:

- Different diseases follow different growth patterns
- Some diseases show stable trends, while others fluctuate significantly
- Seasonal variations are clearly visible in the data

This analysis helps in understanding how diseases evolve over time and supports more accurate forecasting.

4.1.4 Outbreak Detection Results

One of the key features of the system is its ability to detect potential disease outbreaks. The system analyzes predicted values and identifies abnormal increases in disease cases.

The alert dashboard displays:

- Disease name
- Description of the risk
- Predicted number of cases
- Priority level (e.g., High)

The results show that the system successfully identifies high-risk diseases such as:

- Dengue
- Respiratory Infection
- Viral Fever

- Gastritis
- Diabetes

These diseases are flagged with high priority alerts, indicating a significant increase in predicted cases. This allows healthcare providers to take preventive actions and allocate resources accordingly.

4.1.5 Visualization and Dashboard Results

The system provides a modern and interactive dashboard that presents results in an intuitive format. The key visualization components include:

- Line graphs for disease trends
- Forecast charts for future predictions
- Alert tables for outbreak detection
- Summary cards for key metrics

The visualizations enhance the interpretability of the results and allow users to quickly identify important patterns and insights. The use of charts and graphs makes the system more user-friendly and suitable for real-world applications.

4.1.6 Overall System Performance

The overall system performed effectively in terms of:

- Prediction accuracy
- Response time
- User interface functionality

The system successfully integrates machine learning models with a web-based interface, providing real-time predictions and visualizations. The results confirm that the system is capable of supporting hospital decision-making and improving resource management.

4.1 Summary

The results of this research demonstrate that the proposed system is capable of accurately forecasting disease trends and detecting potential outbreaks. The use of machine learning models, combined with visualization tools, provides a powerful solution for healthcare analytics.

The system's ability to generate both short-term and long-term predictions, along with its outbreak detection functionality, makes it a valuable tool for hospitals and healthcare organizations. The selection of Random Forest as the final model ensures high accuracy and reliability in prediction results.

4.2 Research Finding

The implementation and evaluation of the proposed Illness Trend Forecasting and Outbreak Detection system led to several significant findings. These findings highlight the effectiveness of the system in predicting disease trends, identifying outbreaks, and supporting healthcare decision-making. The results obtained from the experiments provide valuable insights into the performance of different models, the impact of input features, and the practical applicability of the system.

4.2.1 Accuracy of Disease Forecasting

One of the key findings of this research is that the system is capable of accurately predicting disease trends using historical data. The predictive models were able to

learn patterns from past disease occurrences and generate reliable forecasts for future time periods.

The results demonstrate that incorporating time-based features such as month and week, along with environmental variables like temperature and humidity, significantly improves the prediction accuracy. The system was able to capture both short-term fluctuations and long-term trends in disease data, making it suitable for healthcare forecasting applications.

4.2.2 Superiority of Random Forest Model

Among all the models tested in this research, the Random Forest model achieved the best performance in terms of accuracy and error reduction. This finding indicates that ensemble machine learning methods are highly effective for handling complex healthcare datasets.

The superior performance of Random Forest can be attributed to its ability to:

- Capture non-linear relationships between variables
- Handle multiple input features effectively
- Reduce overfitting through ensemble learning

Compared to other models:

- LSTM showed moderate performance due to data limitations
- SARIMA was less effective due to its assumption of linear relationships

- Prophet provided reasonable predictions but lacked flexibility for complex patterns

This finding confirms that Random Forest is a suitable model for disease trend forecasting in this context.

4.2.3 Importance of Environmental Factors

Another important finding is the significant impact of environmental factors on disease occurrence. The inclusion of temperature and humidity as input features improved the predictive performance of the models.

The results suggest that:

- Higher humidity levels are associated with increased cases of certain diseases
- Temperature variations influence the spread of respiratory and vector-borne diseases

This finding highlights the importance of incorporating environmental data when developing predictive healthcare systems**.**

4.2.4 Effectiveness of Outbreak Detection Mechanism

The outbreak detection component of the system proved to be effective in identifying abnormal increases in disease cases. The system was able to generate alerts when predicted values exceeded predefined thresholds, indicating potential outbreaks.

The findings show that:

- The system successfully detected high-risk diseases such as dengue and respiratory infections
- Alerts were categorized based on priority levels, improving usability
- Early warnings allow proactive decision-making

This capability enhances the practical value of the system and supports preventive healthcare strategies.

4.2.5 Usability and Visualization Efficiency

The system's user interface and visualization features significantly improved the usability of the application. The dashboards provided clear and intuitive representations of data, enabling users to easily interpret prediction results.

Key observations include:

- Graphical representations helped identify trends quickly
- Dashboard cards provided concise summaries of predictions
- Alert tables clearly highlighted high-risk diseases

This finding emphasizes the importance of visualization in making complex data understandable for non-technical users.

4.2.6 Practical Applicability of the System

The results indicate that the proposed system has strong potential for real-world implementation in healthcare environments. The system can support hospital administrators in:

- Planning resource allocation

- Managing patient inflow
- Responding to disease outbreaks

The use of real-world data from Yakgahapitiya Hospital further strengthens the practical relevance of the system.

4.2.7 Limitations Identified During Research

The study also identified certain limitations that can be addressed in future work:

- The dataset is limited to a single hospital, which may affect generalization
- Lack of real-time data integration reduces responsiveness
- Limited environmental features may restrict prediction accuracy

Recognizing these limitations provides opportunities for further improvement and future research.

4.3 Discussion

The results of this research show that the proposed Illness Trend Forecasting and Outbreak Detection system is effective in predicting disease trends and supporting healthcare decision-making.The system successfully identified patterns in disease data and generated accurate forecasts. Among the models used, Random Forest performed the best, as it was able to capture complex relationships between time, temperature, humidity, and disease occurrence. Other models such as LSTM, SARIMA, and Prophet showed moderate performance but were less accurate for this dataset.The inclusion of environmental factors such as temperature and humidity improved the prediction results. This indicates that external conditions play an important role in disease spread and should be considered in forecasting systems.The outbreak detection feature also worked effectively by identifying diseases with abnormal increases and generating alerts. This helps hospitals take early action and

improve preparedness.The system's dashboards and visualizations made it easy to understand the results. This improves usability and helps non-technical users make better decisions.

However, there are some limitations. The system uses data from only one hospital and does not include real-time data. Adding more data sources and real-time updates could improve the system further.

Overall, the system achieved its objectives and shows strong potential for real-world use in hospital resource planning and disease monitoring.

4.4 Summary of Each Student's Contribution

This research project was conducted as a group effort, where each member contributed to a specific component of the overall Hospital Resource Optimization and Demand Forecasting System.

My Contribution – Illness Trend Forecasting and Outbreak Detection

- Collected and prepared disease dataset from Yakgahapitiya Hospital, Kandy
- Performed data preprocessing and feature engineering
- Implemented multiple prediction models (Random Forest, LSTM, SARIMA, Prophet)
- Compared model performance and selected Random Forest as the best model
- Developed disease forecasting system (weekly and monthly predictions)
- Implemented outbreak detection mechanism with alert system
- Developed web application using React, Node.js, and MongoDB
- Designed dashboards for visualization of predictions and alerts

Adikari A.M.C.G – Staff Attendance Analysis and Patient-Doctor Scheduling

- Developed staff attendance monitoring system
- Implemented patient-doctor scheduling functionality
- Optimized doctor allocation based on availability and demand
- Collected and prepared disease dataset from Yakgahapitiya Hospital, Kandy
- Developed web application using React, Node.js, and MongoDB

Wijethunga S.R – Medicine Demand Forecasting and Accident Surge Detection

- Developed medicine demand prediction model
- Implemented accident surge detection system
- Created alert mechanisms for sudden increases in demand
- Developed web application using React, Node.js, and MongoDB
- Collected and prepared disease dataset from Yakgahapitiya Hospital, Kandy

Lochana J.L.S – Hospital Bed Demand Forecasting and Optimization

- Developed hospital bed demand forecasting model
- Implemented bed allocation and optimization strategies
- Integrated prediction results for efficient resource management
- Collected and prepared disease dataset from Yakgahapitiya Hospital, Kandy
- Developed web application using React, Node.js, and MongoDB

4.4 Summary

Each member contributed to a different functional component, and all modules were successfully integrated to form a complete system. The collaboration ensured that the system addresses multiple aspects of hospital management, including forecasting, resource optimization, and decision support.

3 CONCLUSION

This research presented the design and development of an Illness Trend Forecasting and Outbreak Detection system as part of the broader Hospital Resource Optimization and Demand Forecasting System. The main objective of this study was to develop a data-driven solution capable of predicting disease trends and identifying potential outbreaks to support healthcare decision-making.

The system was successfully implemented using real-world data obtained from Yakgahapitiya Hospital, Kandy. Multiple predictive models, including Random Forest, LSTM, SARIMA, and Prophet, were developed and evaluated. Based on the results, the Random Forest model achieved the highest accuracy and was selected as the final model for disease forecasting.

The system demonstrated the ability to generate both weekly and monthly predictions, allowing hospitals to plan resources effectively. In addition, the outbreak detection mechanism successfully identified abnormal increases in disease cases and generated alerts, enabling early response and improved preparedness.

The integration of environmental factors such as temperature and humidity enhanced the prediction accuracy and allowed the model to capture real-world disease patterns. The web-based implementation using React, Node.js, and MongoDB provided a user-friendly interface with dashboards and visualizations, making it easier for users to interpret results.

Overall, the system achieved its objectives by providing accurate predictions, improving disease monitoring, and supporting hospital resource optimization. The results confirm that machine learning-based approaches can significantly improve healthcare analytics and decision-making.

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APPENDICES

Appendix A: Frontend System Interface

Figure A.1: Alert Page

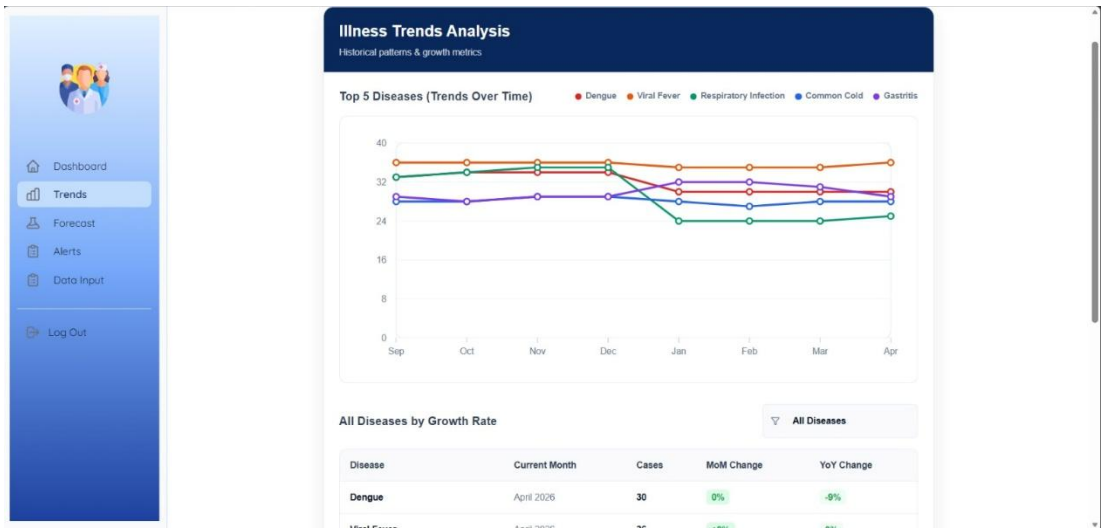

Figure A.2: Illness Trend Analysis

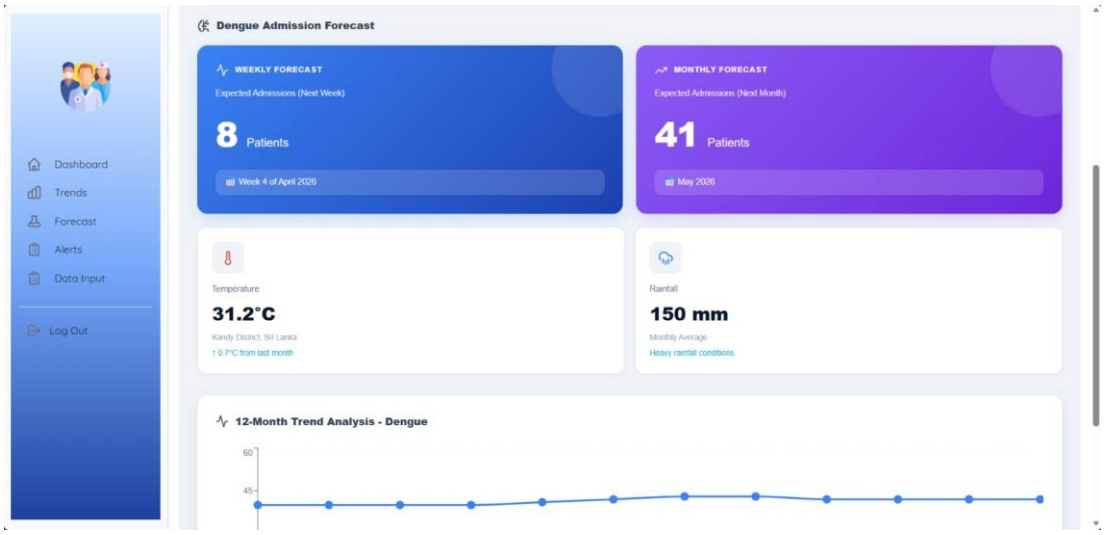

Figure A.3: Dashboard

Figure A.4: Forecasting Page

Appendix B : System Frontend Codes

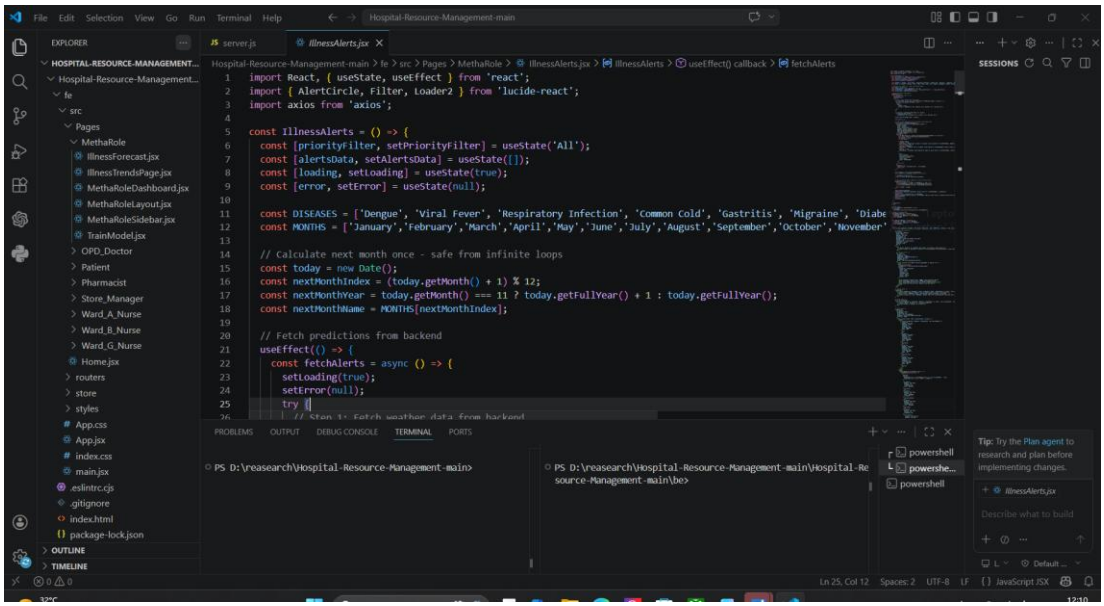

Figure B.1: Frontend (Alert Page)

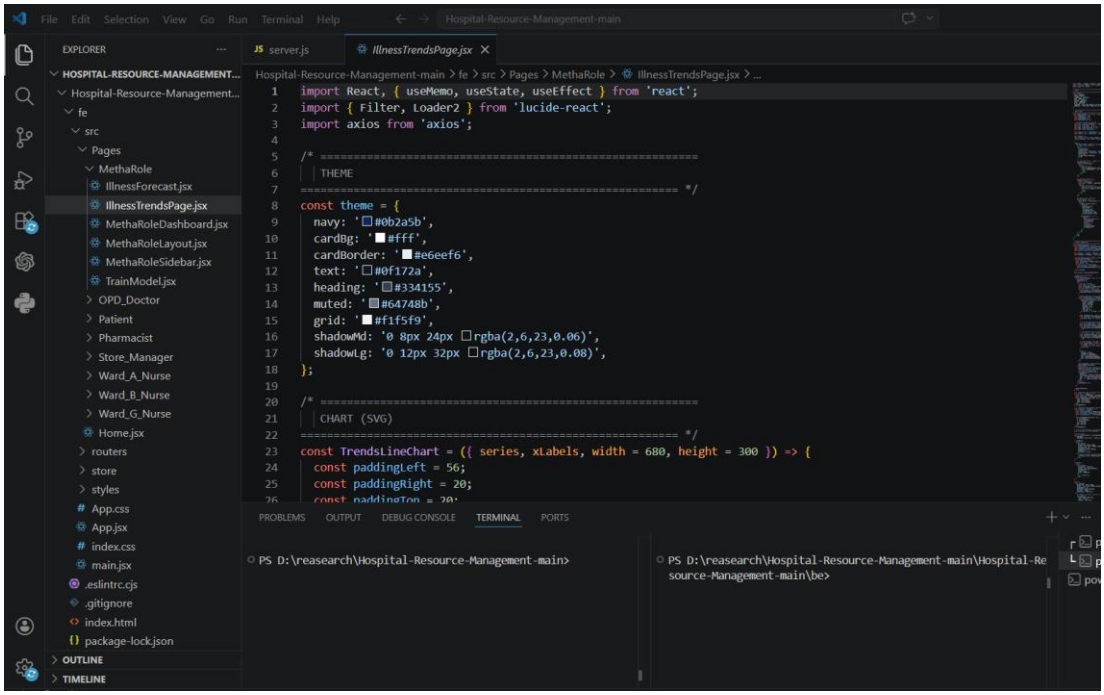

Figure B.2: Frontend(Illness trend page)

Appendix C: Backend Model Training (Random Forest)

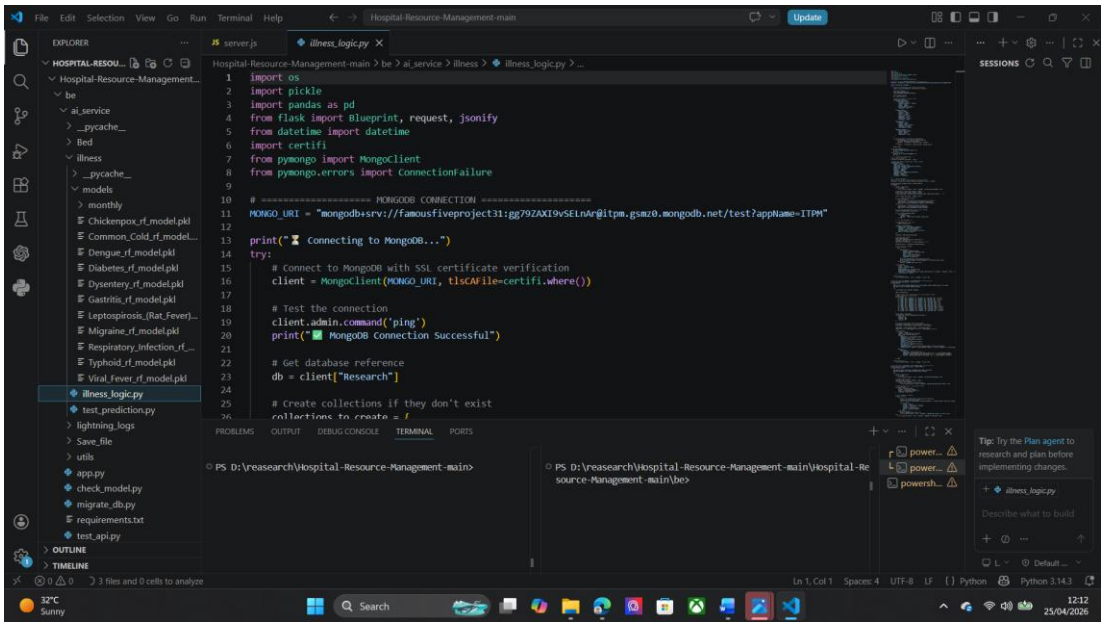

Figure C.1: Model Training Connection For Backend(Random Forest)